ISB 120

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Unit 2: Supply and Demand Analysis

- The law of demand: factors influencing consumer demand
- The law of supply: factors influencing producer supply
- Market equilibrium and price determination: the interplay of supply and demand
- Shifts in supply and demand curves: causes and consequences
- Price elasticity of demand and supply: measuring responsiveness to price changes
- Cross-price elasticity and income elasticity
- Price controls and their impacts on markets
- Case studies of supply and demand in specific industries
- Government intervention in markets and its effects
- Global supply and demand considerations and international trade implications

The law of demand: factors influencing consumer demand

- In general, Demand means desire by human.
- In economics, Demand refers to the desire backed by ability to pay and willingness to buy it.
- A beggar may desire to have a car, but his desire is not going to affect its market price as he is not having the necessary purchasing power to buy a car.
- Thus, desire backed by purchasing power is called demand.

KINDS OF DEMAND

I. Direct Demand: It refers to demand for a commodity that is directly consumed to satisfy human wants, for example demand for bread, butter and fruits.

a) **Price Demand:** it refers to the demand for a commodity at a particular price

b) **Income Demand:** It refers to the demand for a commodity at a various levels of consumer's income

c) **Cross Demand:** It refers to quantity demanded of a commodity due to change in the price of other commodity

II. Indirect Demand or Derived Demand:

- Demand for factors of production is indirect because they help in the production of a commodity which is directly demanded by the consumer in the market.

III. Complementary Demand/ Joint Demand:

- It refers to the demand for those goods for a commodity which are always demanded jointly. Example: car and petrol.

IV. Composite Demand:

- It refers to the total demand for a commodity which can be used for various purposes.

LAW OF DEMAND – ALFRED MARSHALL

- Definition: The law of demand states that , other things remaining equal,
- The quantity demanded for a commodity increases when its price falls and
- Decreases when the price rises.
- There is a inverse relationship between the price of the good and the quantity demanded of that good.

FACTOR AFFECTING THE DEMAND

1. Price of the Given Commodity: It is the most important factor affecting demand for the given commodity. Generally, there exists an inverse relationship between price and quantity demanded.

2. Price of substitutes/related goods: Some goods can be substituted for other goods. For example, tea and coffee are substitutes. If the price of coffee increases while the price of tea remains the same, there will be increase in the demand for tea and decrease in the demand for coffee.

3. Income of the consumer: When the income of the consumer increases, more will be demanded. Comforts and luxuries belong to this category

4. Tastes and preferences of the consumer: Demand for a commodity may change due to a change in tastes, preferences and fashion.

5. Expectation of future price change: If the consumer believes that the price of a commodity will rise in the future, he may buy a larger quantity in the present. Suppose he expects the price to fall, he may defer some of his purchases to a future date.

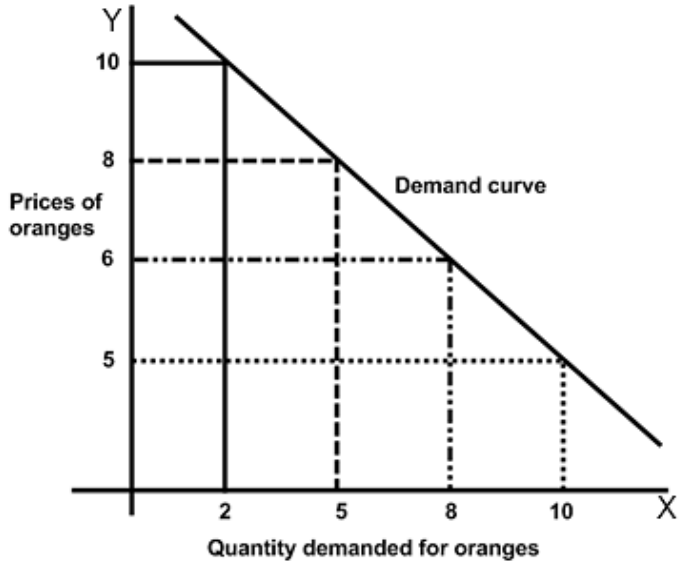
ASSUMPTIONS OF THE LAW OF DEMAND

- The price of the related goods remains the same
- The income of the consumers remain unchanged
- Tastes and preferences of the consumer remain the same

DEMAND SCHEDULE FOR ORANGE

Price of Oranges (Kwacha)	Quantity of Oranges
10	2
8	5
6	8
5	10

DEMAND CURVE



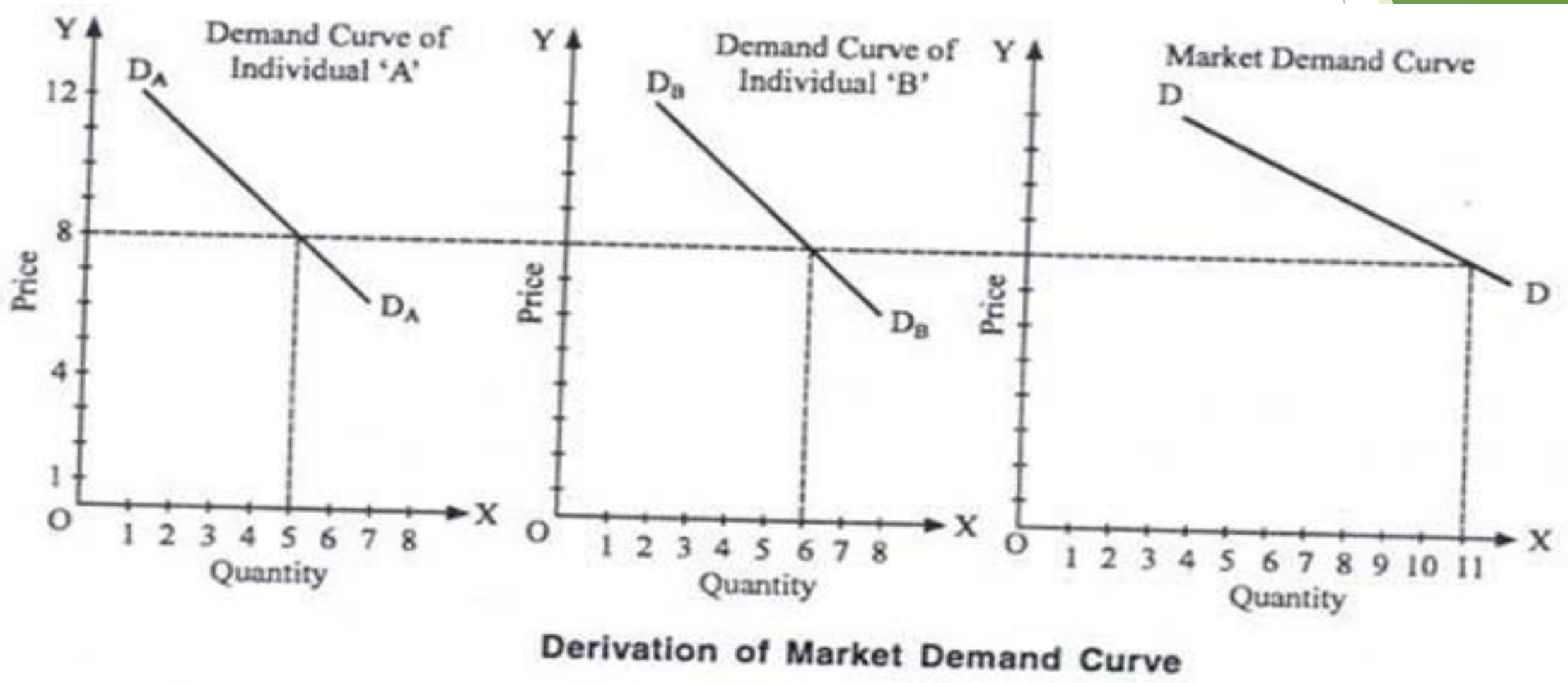
MARKET DEMAND SCHEDULE

- A demand schedule for a market can be constructed by adding up demand schedules of the individual consumers in the market.
- Suppose that the market for Apple consists of 2 consumers. The market demand is calculated as follows.

MARKET DEMAND SCHEDULE FOR APPLE

Price (Kwacha)	Quantity Demanded		
	Quantity A	Quantity B	Quantity A+B
12	1	2	3
11	2	3	5
10	3	4	7
9	4	5	9
8	5	6	11

DERIVATION OF MARKET DEMAND CURVE



Derivation of Market Demand Curve

REASONS BEHIND DOWNWARD SLOPE OF THE DEMAND CURVE

1. Law of diminishing marginal utility: The law of demand is based on the law of diminishing marginal utility which states that as the consumer purchases more and more units of a commodity,

- The utility derived from such successive unit goes on decreasing.
- Like that, consumer purchases more of the commodity so that his marginal utility from the commodity falls to be equal to the reduced price and vice- versa.

2. Substitution Effect: The substitution effect is the effect that a change in relative prices of substitute goods changes the quantity demanded.

- When the price of a good rises the consumer prefers to buy its substitute goods which have became relatively cheaper.

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- **3. Income Effect:** Change in demand on account of change in real income resulting from change in the price of a commodity is known as income effect.
- **4. Several Uses.** Some commodities can be put to several uses which lead to downward slope of the demand curve.
- When the price of such commodities goes up they will be used for important purposes, so their demand will be limited.

EXCEPTIONS TO THE LAW OF DEMAND

- The Law of demand is a general statement telling that prices and quantities of a commodity are inversely related.
- There are certain peculiar cases in which the law of demand will not hold good.
- **I. Veblen Effect:** Veblen has pointed out that there are some goods demanded by very rich people for their social prestige.
- When price of such goods rise, their use becomes more attractive and they are purchased in larger quantities.
- Demand for diamonds from the richer class will go up if there is increase in price. If such goods were cheaper, the rich would not even purchase.

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- **II. Giffen Paradox :** Sir Robert Giffen discovered that the poor people will,
- Demand more of inferior goods if their prices rise and
- Demand less if their prices fall.
- For example, poor people spend the major part of their income on coarse grains (e.g. Maize) and only a small part on rice.

MOVEMENT IN DEMAND CURVE OR EXPANSION AND CONTRACTION OF DEMAND

- A movement along the demand curve is caused by a change in the price of the goods only other things remaining constant.

I. Expansion of Demand:

- It refers to rise in demand due to fall in the price of the goods.

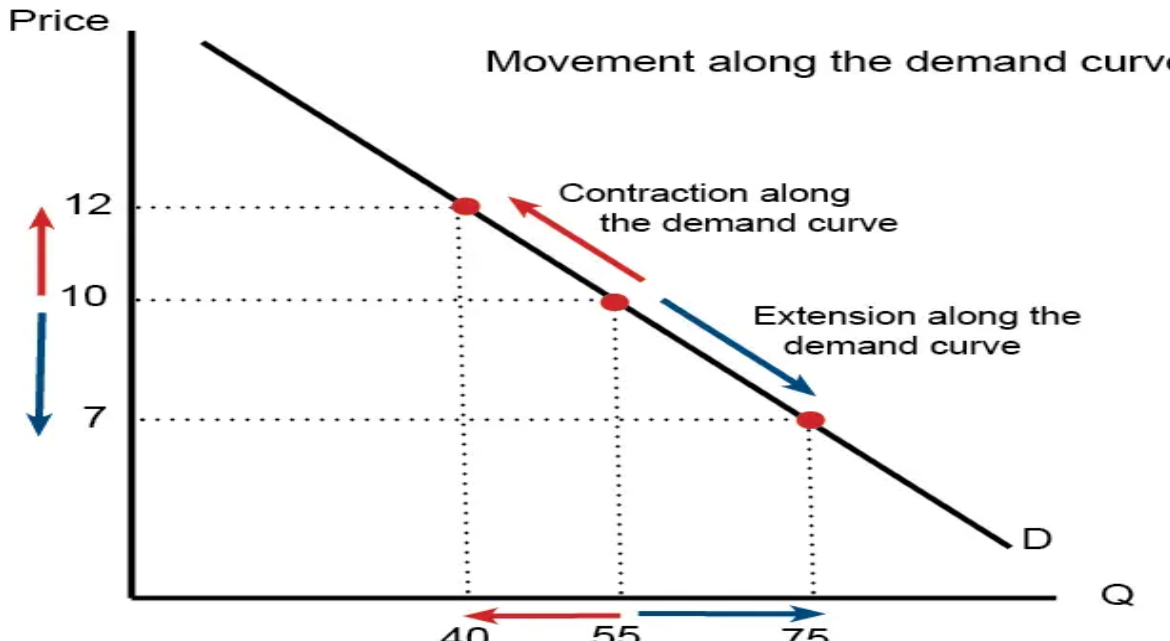
II. Contraction of Demand:

- It refers to fall in demand due to rise in the price of goods.

MOVEMENT IN DEMAND CURVE OR EXPANSION AND CONTRACTION OF DEMAND

Expansion in demand: A fall in price from K16 to K12 leads to an expansion (increase) in demand. As price falls, there is a movement along the demand curve and more is bought.

Contraction in demand: An increase in price from K12 to K16 causes a movement along the demand curve, and quantity demand falls from 80 to 60. We say this is a contraction in demand



SHIFTS IN DEMAND OR INCREASE AND DECREASE IN DEMAND

The shift of the demand curve is caused by changes in factors other than price of goods. These factors are

A) Consumer's income

B) Price of relative or substitute goods

C) Consumer's taste and preferences

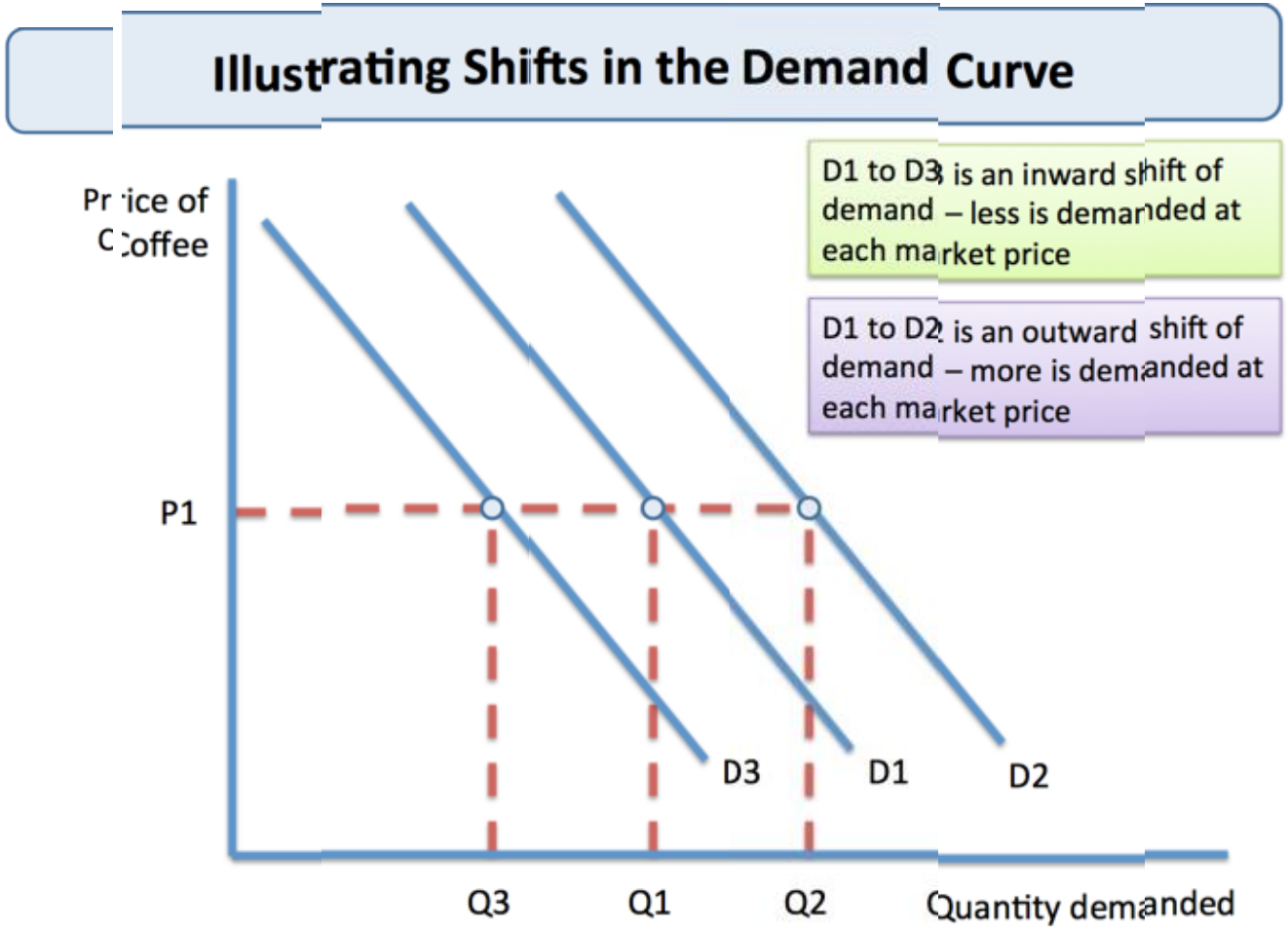
I. Increase in Demand:

- It refers to the situation when the consumers buy a large amount of commodity at the same price.
- The reasons are;
- Increases in the income of consumer
- Increase in the price of substitute goods
- Expectation of rise in price in future
- Increase in population

II. Decrease in Demand:

- It refers to a situation when the consumers buy a smaller quantity of thee commodity at the same price.
- The reasons are;
- Fall in the income of thee consumers
- Fall in the price of the substitute goods
- Expectation of fall in price
- Decrease in Population
- Consumers' taste becoming unfavorable towards the goods

SHIFTTS IN DEMMAND OR IINCREASEE AND DECREASE IIN DEMANND



ELASTICITY OF DEMAND - ALFRED MMARSHAALL

- The law of demand explains that demand will change due to the change in the price of the commodity
- But does not explain the rate at which demand changes due to changes in price,
- The concept of elasticity of demand measures the rate of change in demand
- **DEFINITION OF ELASTICITY OF DEMAND:**
- According to him “the elasticity (or responsiveness) of demand in a market is great or small according as the amount demanded increases much or little for a given fall in price, and diminishes much or little for a given rise in price”.

FACTORS DETERMINING THE PRICE ELASTICITY OF DEMAND

- **1. Availability of Substitute:** Goods having close substitutes will have an elastic demand and goods with no close substitutes will have an inelastic demand. Commodities such as Pen, Pepsi, Maruti car have close substitutes and hence have an elastic demand.
- **2. Income of the consumers:** If the income level of consumers is high, the elasticity of demand will be less. It is because change in the price will not affect the quantity demanded by greater proportion
- **3. Luxuries versus Necessities:** The price elasticity of demand is likely to be low for necessities and high for luxuries
- **4. Number of uses of the commodity:** The more the number of uses of a commodity has more elastic demand. If a commodity has few uses it has an inelastic demand.
- **5. Cost relative to total income:** higher the cost of the goods relative to total income of consumer more will be the price elasticity demand.
- **6. Level of price:** if the price of the commodity is high the price elasticity of demand is more and if it is low, its price elasticity of demand is less.

TYPES OF ELASTICITY OF DEMAND

- Price Elasticity of Demand
- Income Elasticity of Demand
- Cross Elasticity of Demand
- Advertising elasticity of Demand
- Elasticity of Price Expectations

PRICE ELASTICITY OF DEMAND

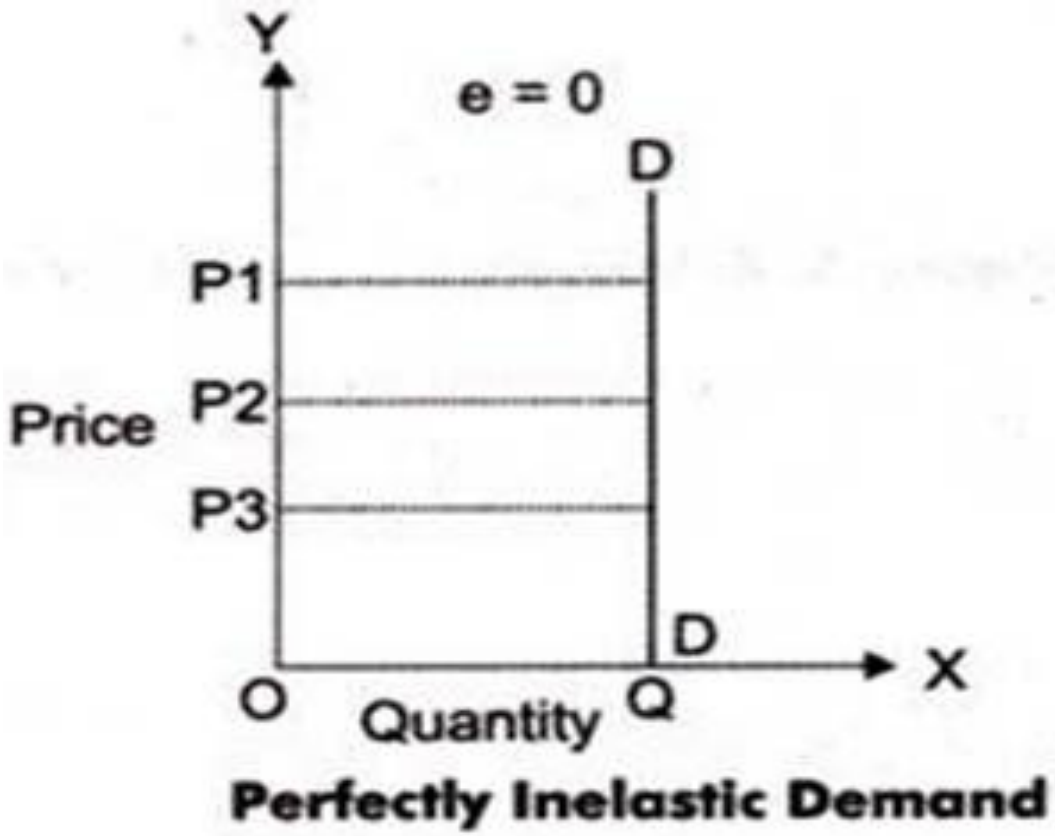
"The degree of responsiveness of quantity demanded to a change in price is called price elasticity of demand"

$$Price\ Elasticity\ of\ Demand = \frac{Percentage\ Change\ in\ Quantity(\Delta Q/Q)}{Percentage\ Change\ in\ Price(\Delta P/P)}$$

DIFFERENT TYPES OF PRICE ELASTICITY OF DEMAND

- **1. PERFECTLY INELASTIC DEMAND (EP = 0):**
- **Description:** when to a percentage change in price there is no change in quantity demanded.
- **Types of Goods:** Essentials like life saving goods
- Changes in price yield no change in demand

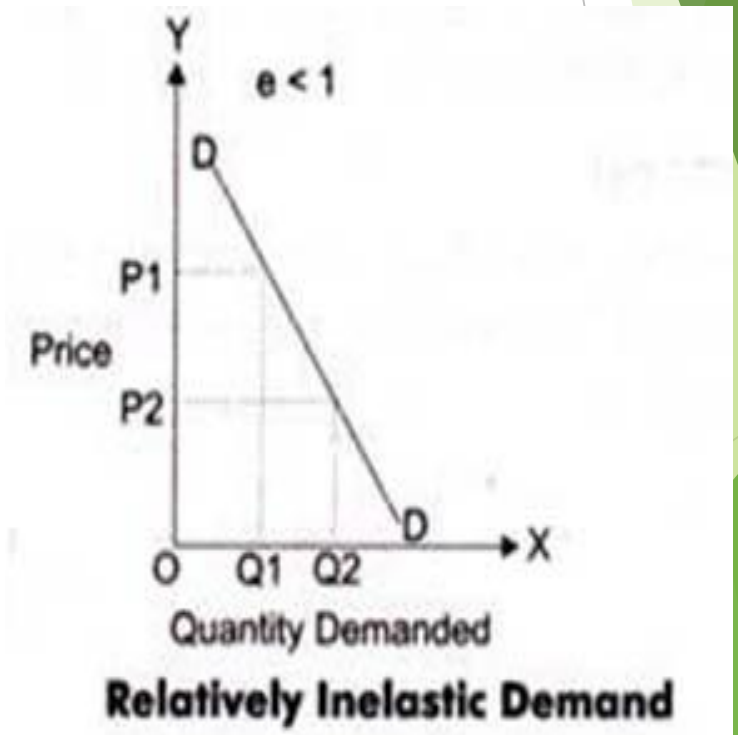
Shape of Demand curve: Vertical



Perfectly Inelastic Demand

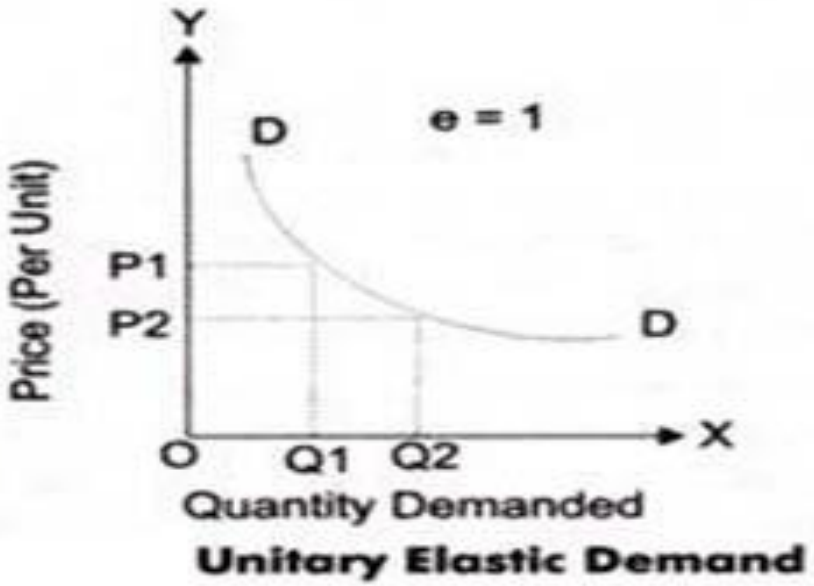
2. RELATIVELY INELASTIC DEMAND (EP< 1)

- **Description:** when to a percentage change in price there is less than proportionate change in quantity demanded.
- **Types of Goods:** Necessities like food, fuel
- **Shape of Demand curve:** Steeper
- Changes in price yield an insignificant change in demand



3. UNITARY ELASTIC DEMAND (EP = 1)

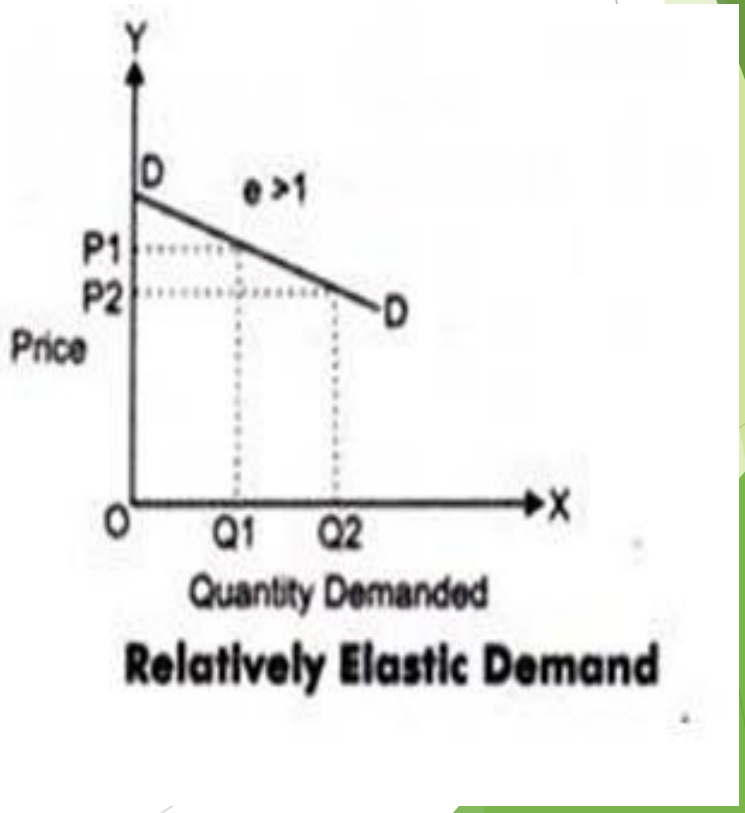
- **Description:** when to a percentage change in price there is equal change in quantity demanded.
- **Types of Goods:** Normal goods
- **Shape of Demand curve:** the linear demand curve forming 45o angle both the axes
- Changes in price yield equivalent (%) changes in demand



Unitary Elastic Demand

4. RELATIVELY ELASTIC DEMAND (EP> 1)

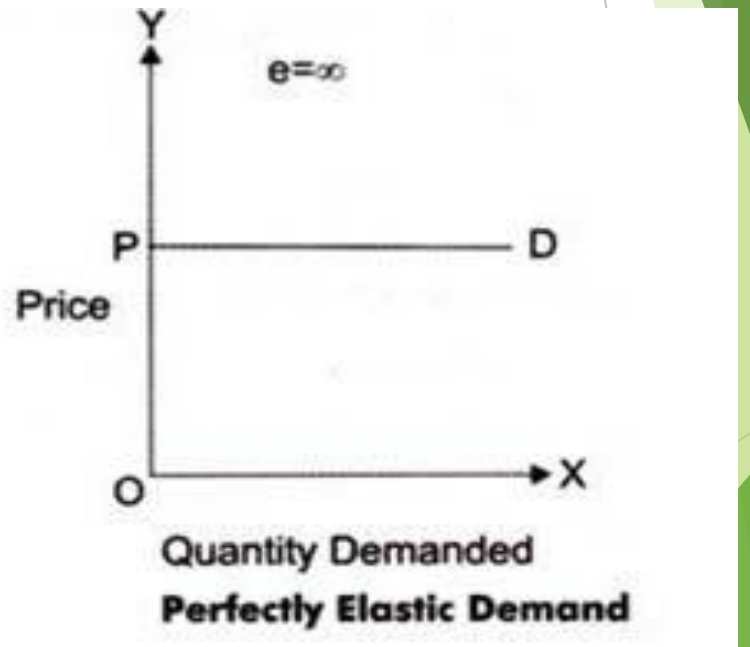
- **Description:** when to a percentage change in price there is more than proportionate change in quantity demanded.
- **Types of Goods:** Luxuries
- **Shape of Demand curve:** Flatter
- Changes in price yield a significant change in demand



Relatively Elastic Demand

5. PERFECTLY ELASTIC DEMAND (EP = ∞)

- **Description:** when there is infinite change in quantity situation demanded without any changes in price
- **Types of Goods:** Imaginary
- **Shape of Demand curve:** Horizontal
- Changes in price result in demand declining to zero



METHODS OF CALCULATING PRICE ELASTICITY OF DEMAND

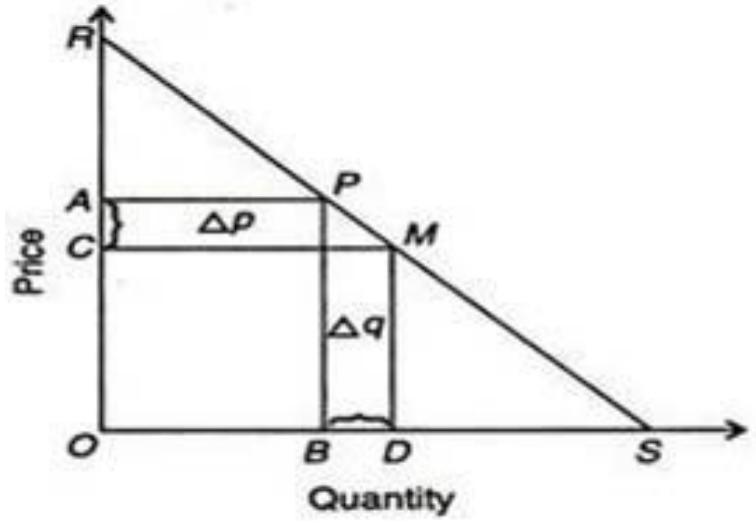
- **1. The Percentage Method:**
- The price elasticity of demand is measured by its coefficient (Ep). This coefficient (Ep) measures the percentage change in the quantity of a commodity demanded resulting from a given percentage change in its price.

$$E_p = \frac{\%\ change\ in\ q}{\%\ change\ in\ p} = \frac{\Delta q / q}{\Delta p / p} = \frac{\Delta q}{\Delta p} \times \frac{p}{q}$$

Where q refers to quantity demanded, p to price and Δ to change. If $E_P > 1$, demand is elastic. If $E_P < 1$, demand is inelastic, and $E_p = 1$, demand is unitary elastic.

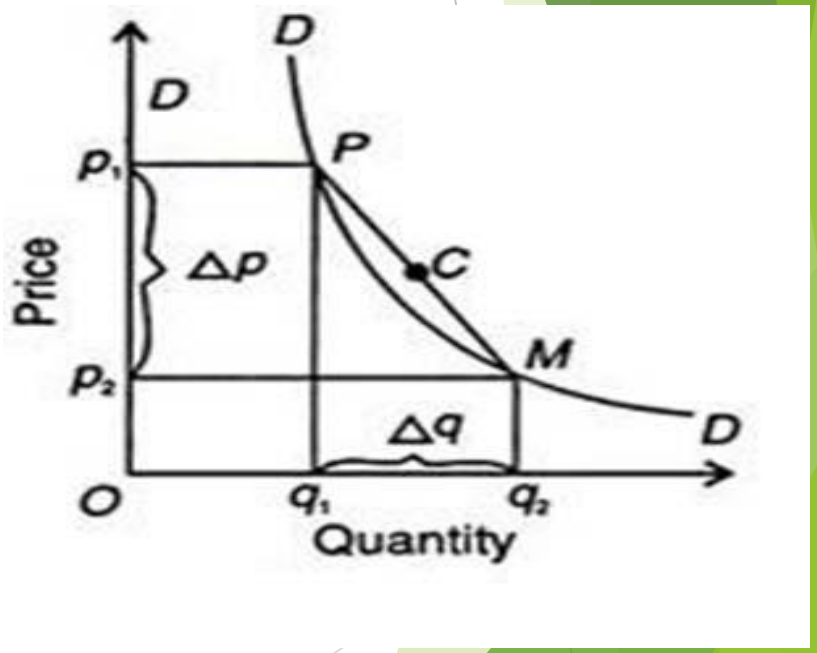
2. The Point Method or Geometrical or graphical method:

- The point method of measuring elasticity of demand was developed by Alfred Marshall. Elasticity measures at a point on a demand curve is known as point elasticity of demand
- Let RS be a straight line demand curve in Figure. If the price falls from PB (= OA) to MD (= OC), the quantity demanded increases from OB to OD.



3. The Arc Method:

- When elasticity of demand is measured over a finite range or ‘arc’ of a demand curve, it is called arc elasticity of demand. when elasticity is measured between two points on the same demand curve, it is known as arc elasticity.



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- The area between P and M on the DD curve in Figure is an arc which measures elasticity over a certain range of price and quantities.
- Elasticity for the arc (PM in Figure) is calculated by taking the average of the two prices [(p1 + p2)½] and the average of the two quantities [(q, +q2)½].
- The formula for price elasticity of demand at the mid-point (C in Figure 4) of the arc on the demand curve.

The Total Outlay Method:

- Marshall evolved the total outlay, or total revenue or total expenditure method as a measure of elasticity. By comparing the total expenditure of a purchaser both before and after the change in price, it can be known whether his demand for a good is elastic, unity or less elastic.
- Total outlay is price multiplied by the quantity of a good purchased: Total Outlay = Price x Quantity Demanded.

EXAMPLES

- A dealer in deep freezers increased the price of his products from K450 to K500.
- And his sales dropped from 800 units to 600 units,
- Use the information above to answer the questions below:
- A) Calculate the price elasticity of demand
- B) What type of elasticity is this?
- C) Was the decision to increase price Okay for the business? Why?
- P1=K450, P2 = K500, Change in P, =50
- Q1 = 800 units, Q2 = 600 units Change in Q = -200 units
- Ep = (Change in Q)/Q x P/(Change in P) = -200/800 x 450/50 = -2.25

CONT…

- Or % Change in Qd /% Change in P = [((600-800)/800)x100]/[((500-450)/450)x100] = 25% /11.1% = 2.25
- The elasticity is Relative Elastic demand

- Q2. Calculate the arc elasticity using the data below
- P1 = K1000, P2 = 800 Q1 = 200 UNITS , Q2 = 300 UNITS
- Pe = [(Q2 – Q1)/9(Q1+Q2)/2] / [(P2-P1)/(P1+P2)/2]
- Pe = - 1.8

INCOME ELASTICITY OF DEMAND

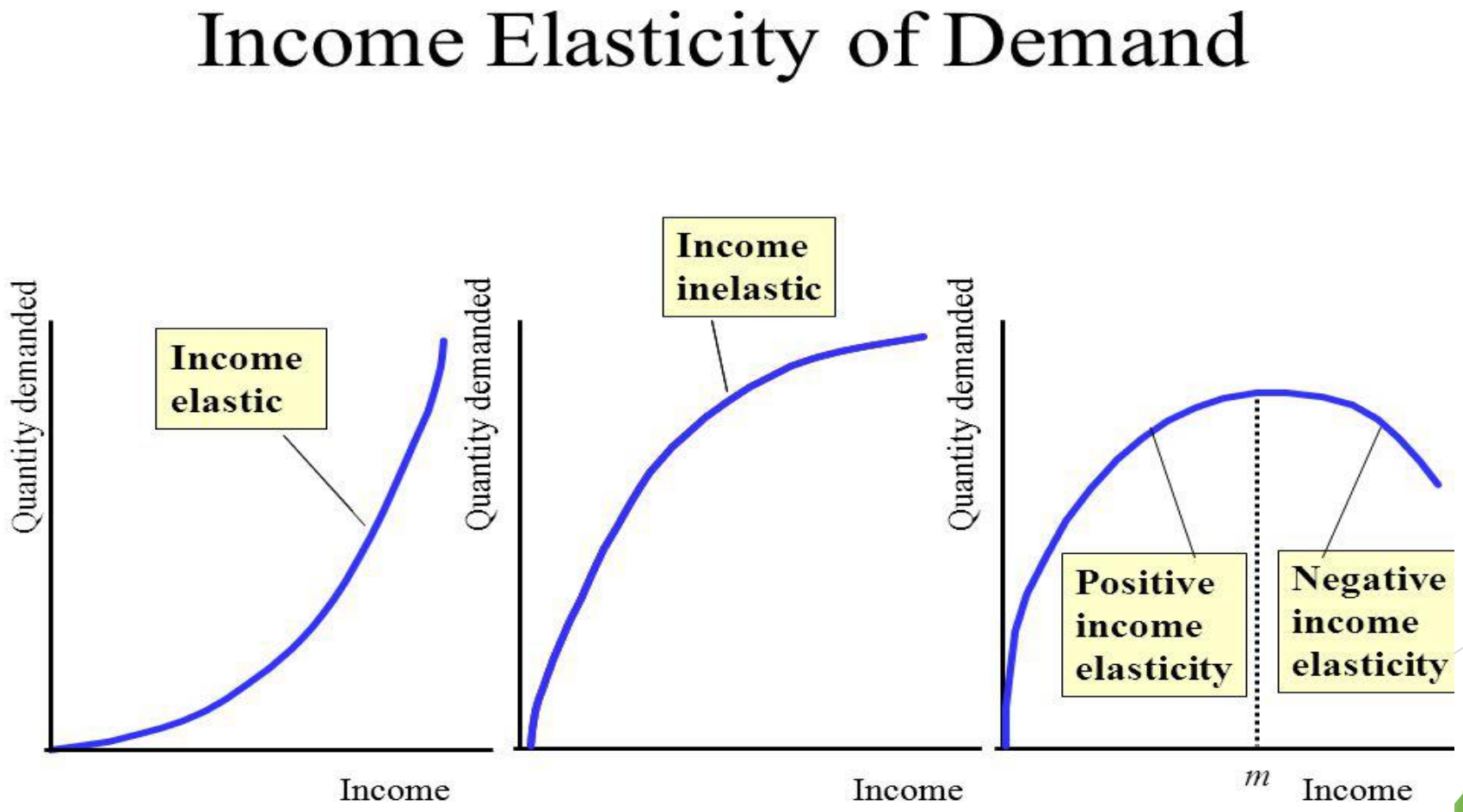
- Income elasticity of demand is the degree of responsiveness of demand to the change in income.

$$E_y = \frac{\text{Percentage change in the qsuantity demanded}}{\text{Percentage change in income}}$$

$$\frac{\Delta Q/Q}{\Delta Y/Y} = \frac{\Delta Q}{Q} \times \frac{Y}{\Delta Y} = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

- Where,
- Q = Original quantity
- Y = Original income
- ΔQ = Changes in quantity
- ΔY = Changes in income
- Ey = Income elasticity of demand

CONT…


Income Elasticity of Demand
Quantity demanded
Income elastic
Income
Quantity demanded
Income inelastic
Income
Quantity demanded
Positive income elasticity
Negative income elasticity
m
Income

CROSS ELASTICITY OF DEMAND

Cross Elasticity of Demand measures the responsiveness of the change in quantity demanded of one commodity due to a change in the price of another commodity. The degree of responsiveness of quantity demanded of one commodity to changes in the price of another commodity is called cross elasticity of demand.

$$EC = \frac{\text{Percentage change in quantity demanded of Good} - X}{\text{Percentage change in the price of Good} - Y}$$

$$= \frac{\dfrac{\text{Change in quantity demanded of X}}{\text{Original Quantity of X}}}{\dfrac{\text{Change in Price of Y}}{\text{Original Price of Y}} \times 100} \times 100$$

$$= \frac{\dfrac{\Delta Q_x}{Q_x}}{\dfrac{\Delta Q_y}{Q_y}} = \frac{\Delta Q_x}{Q_x} \times \frac{P_y}{\Delta P_y}$$

$$EC = \frac{P_y}{\Delta_n} \times \frac{\Delta Q_n}{\Delta P_y}$$

Where

Py = Original price of good-Y

ΔPy = Change in price of good-Y

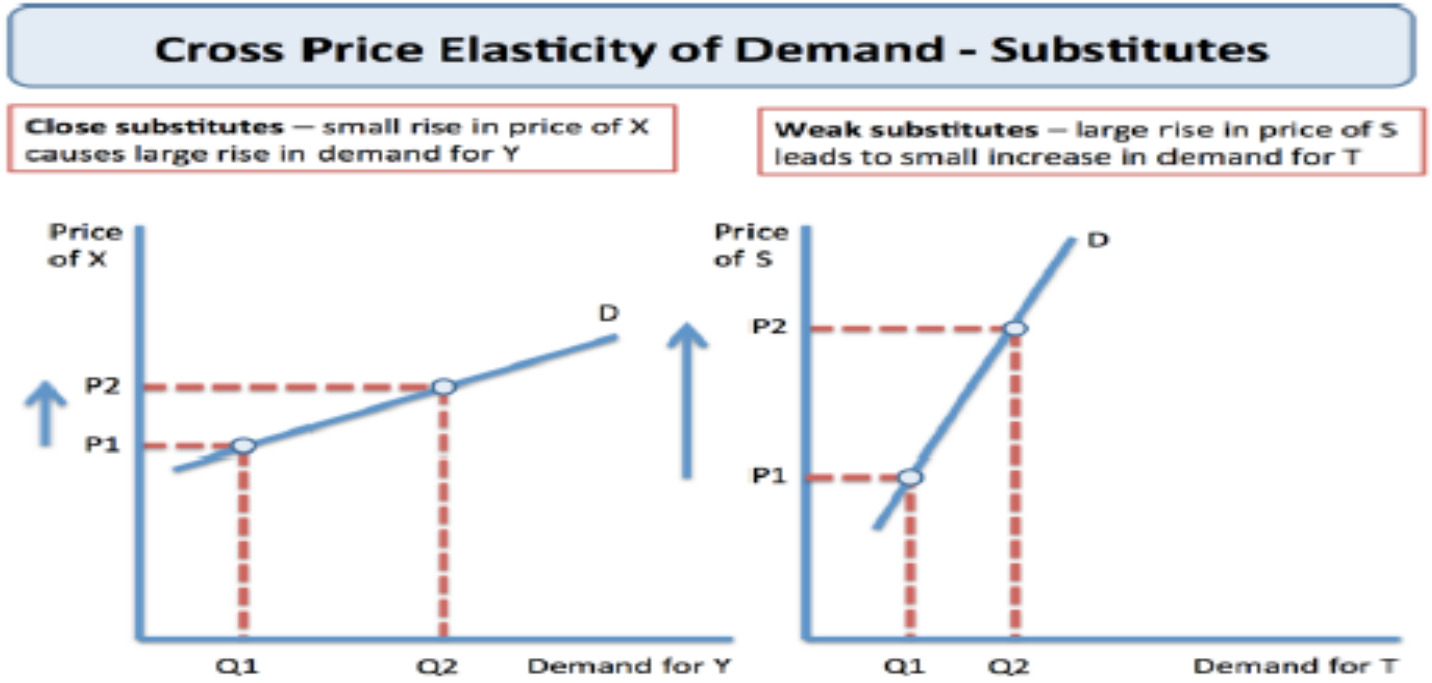
Q_x = Original quantity demanded of X

ΔQ_x = Change in the quantity demanded of X

- Cross elasticity of demand varies from minus infinity to plus infinity. Complementary goods have negative cross elasticity and substitute goods have positive cross elasticity.

CONT…

- Complement goods like bread and butter, bricks and cements, pen and ink will negative cross elasticity
- Substitute goods like coffee and tea have positive cross elasticity



ADVERTISING ELASTICITY OF DEMAND

- It measures the response of quantity demanded due to change in advertising expenditure. It is also called promotional elasticity of demand

$$E_a = \frac{\text{Proportionate change in Demand}}{\text{Proportionate change in Advertising Expenditure}}$$

Where: Q= Initial Quantity Demanded

ΔQ= Change in quantity demanded

A= Initial advertising expenditure

ΔQ = Change in advertising expenditure

e_A = Coefficient of advertising elasticity of demand

DETERMINANTS OF ADVERTISING ELASTICITY OF DEMAND

The Main factors affecting advertising elasticity of demand are:

1. Stage of product market: Advertising elasticity is different for new products and for old products.

- It is also different for products with established market and a growing market.

2. Effect of advertising by rivals: The effectiveness of advertising depends upon how the rivals react to the advertising campaign of this firm.

3. Effect of advertising in terms of time: the time lag in response to advertisement differs.

- It depends upon the types of product. It takes longer in case of durable good because will buy durable goods only after the existing one has been used up.

ELASTICITY OF PRICE EXPECTATIONS

- The concept of elasticity of price expectations was developed by J.R.Hicks.
- Elasticity of price expectations is defined as the ratio of the relative change in expected future prices to the relative change in current price.
- It is symbolically
- =

$$ep_e = \frac{\%\text{ change in expected future price}}{\%\text{ change in current price}} \qquad \frac{\Delta E_p}{\Delta C_p} \times \frac{E_p}{C_p}$$

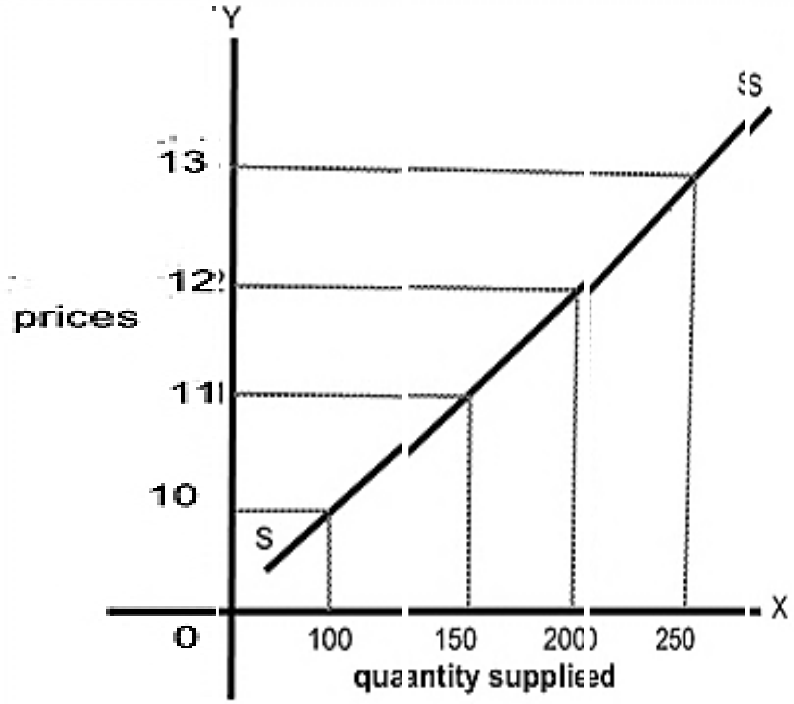
LAW OF SUPPLY

- The relationship between price and quantity supplied is usually a positive relationship. A rise in price is associated with a rise in quantity supplied.
- **Definitions**
- **Dooley says** "The law of supply states that other things being equal the higher the price, the greater the quantity supplied or the lower the price, the smaller the quantity supplied.“
- **Lipsey, defines it as "The** law of supply states that other things being equal, the quantity of any commodity that firms will produce and offer for sale is positively related to the commodity's own price, rising when price rises and falling when price falls.“
- As the price of good increases, suppliers will attempt to maximize profits by increasing the quantity of the product sold.

SUPPLY SCHEDULE

PRICE OF APPLES	QUANTITY OF APPLES SUPPLIED
10	100
11	150
12	200
13	250

SUPPLY CURVE



DETERMINANTTS OF SUPPLY

1. Cost of factors of production

- Price of raw materials
- Rent and interest on Machinery
- Cost of Machinery
- Payments on Human Resources (wages & Salaries)
- Payments on Transport
- If Production is high Supply will be low

2. State of technology

- Use of technology reduces cost of production
- Increases production capacity
- Which increases supply

3. Factors outside the Economic Sphere.

- The factors below should not rise, if they rise they affect supply directly or indirectly.
- Weather conditions,
- Wars
- Unexpected situations like epidemics (COVID-19)
- Floods

5. TAX AND SUBSIDY

- If Tax subsidy (Charge less tax) is given by the government
- Cost of production is low but if not
- The cost of production rises
- Leading to low production which will effect decrease in supply.

6. Goal of business firm

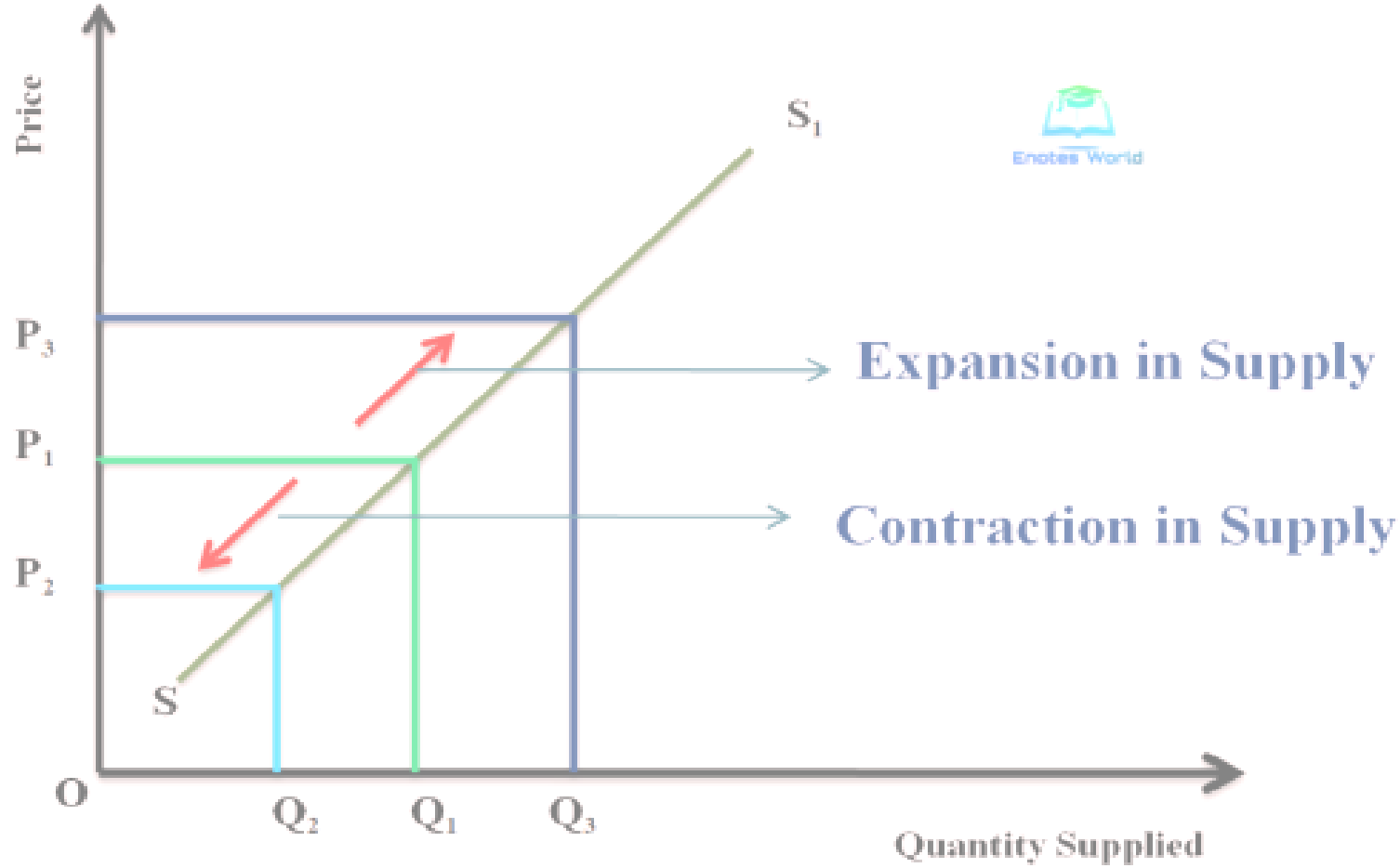
7. Price of related goods

8 Change in number of companies in the industry

Movement along the Supply Curve

- A change in price causes a movement along the supply curve; such a movement is called a change in quantity supplied.
- As is the case with a change in quantity demanded, a change in quantity supplied does not shift the supply curve. By definition, it is a movement along the supply curve.
- For example, if the price rises from P1 to P3, the quantity supplied rises from Q1 to Q3 and that is an expansion in supply.
- When the price drops from P1 to P2, the quantity supplied drops from Q1 to Q2 and that is a contraction in supply.

SUPPLY CURVE



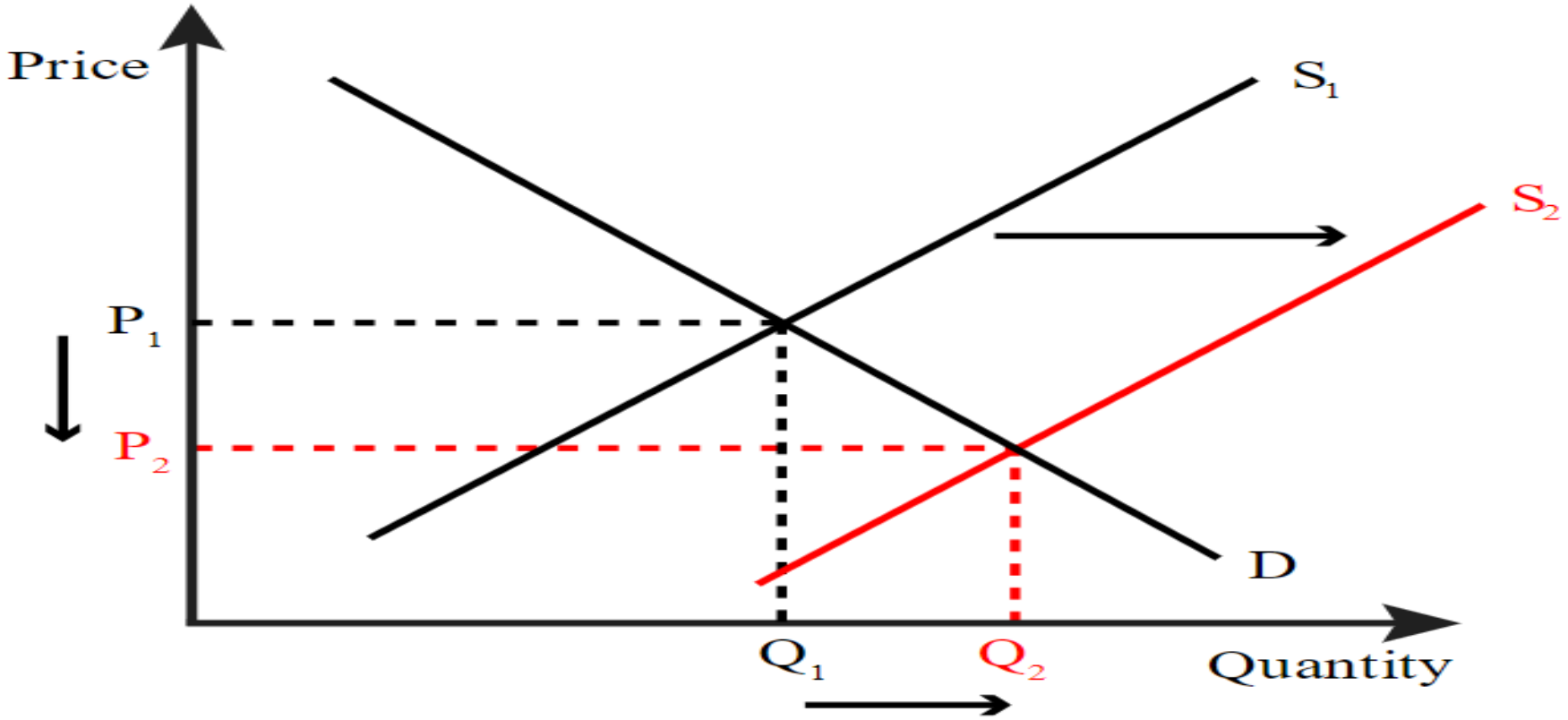
Shifts in the Supply Curve

- Shifts in supply curve arise from changes in the following
- **The cost of production for the firms:**
- For example, autoworkers in Zambia represented by a union might negotiate a new labour contract for higher wages.
- This represents a cost increase on the part of firms and will result in a leftward shift of the supply curve.
- **Technology that affects the production function:**
- For example, a new robotic technology that aids in the assembly of automobiles in the factory could improve the efficiency of the plant, resulting in a rightward shift of the supply curve.

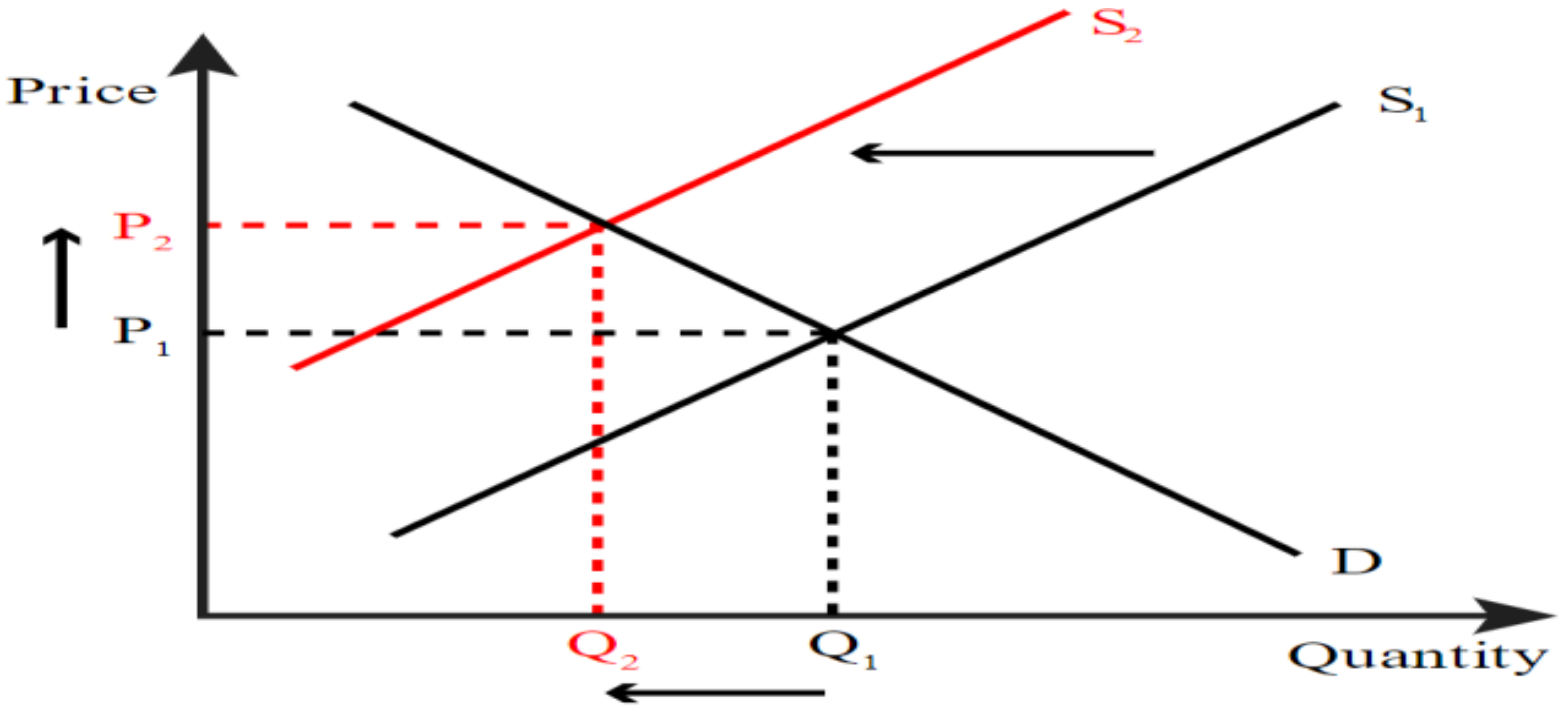
SHIFTS IN SUPPLY CURVE

- **The number of sellers in the market:**
- For example, perhaps a new company, like Toyota decides to begin to manufacture cars in Zambia, the number of sellers has increased, and the number of cars for sale is likely to increase as well, leading to a rightward shift in the supply curve.
- ▫ **The presence of alternative markets for sellers:**
- For example, an economic boom in Zambia might cause South African manufacturers to send more of the cars they produce to Zambia instead of selling them in South African, leading to a lower supply of cars in the South African market represented by a leftward shift in the supply curve.

Increase in Supply



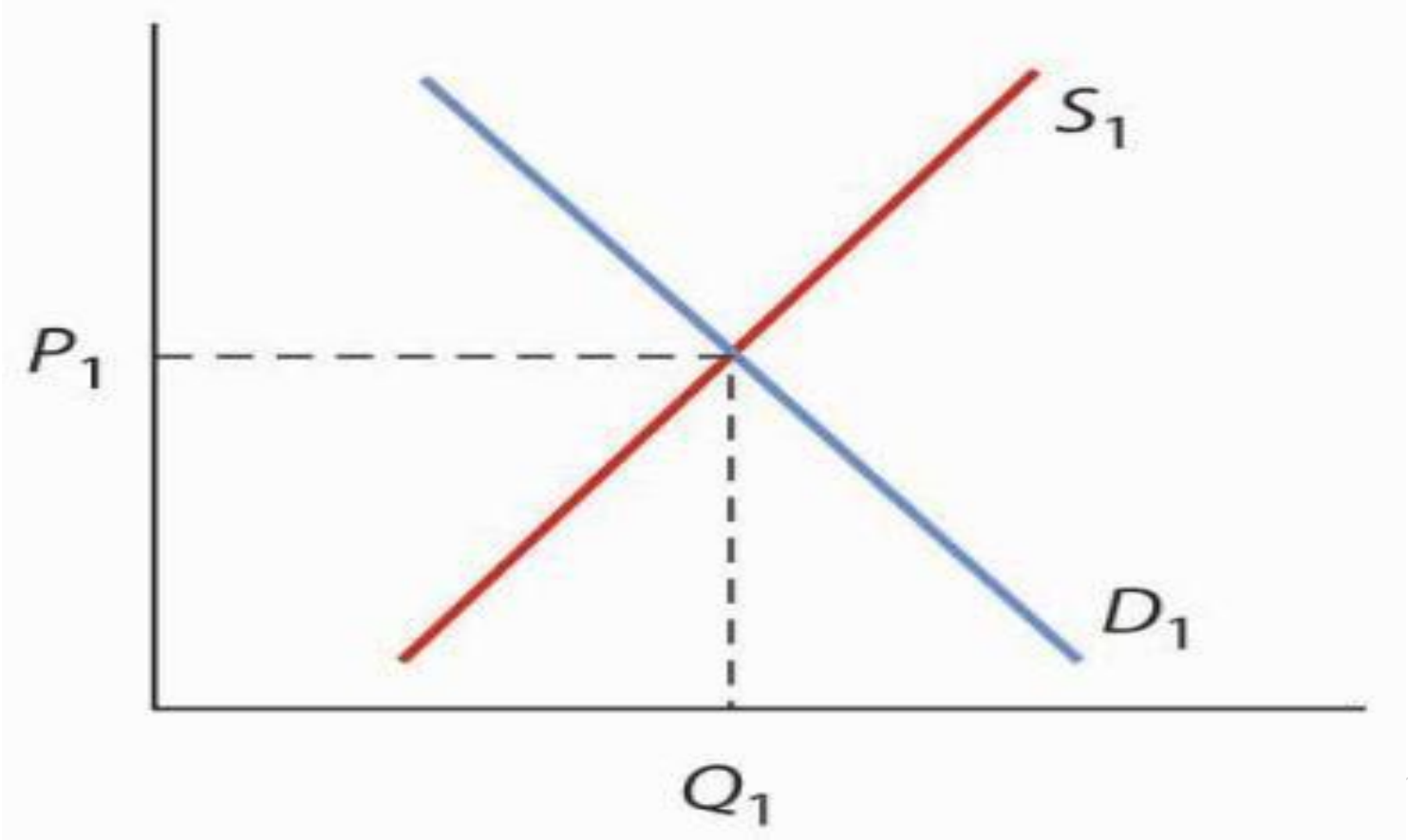
Decrease in Supply



Equilibrium Demand and Supply, Consumer Surplus and Producer surplus

- Market equilibrium is the quantity-price point where supply and demand balance out in such a way that quantity demanded equals quantity supplied.
- The market stabilizes at the price that corresponds to this quantity. When visualized on a graph, the economic equilibrium is the point of intersection between the demand and supply curves.
- This point of intersection reflects the compromising quantity that both consumers are willing to seek out and producers are willing to supply at a certain price.

EQUILIBRIUM CURVE



Market Supply Schedule:

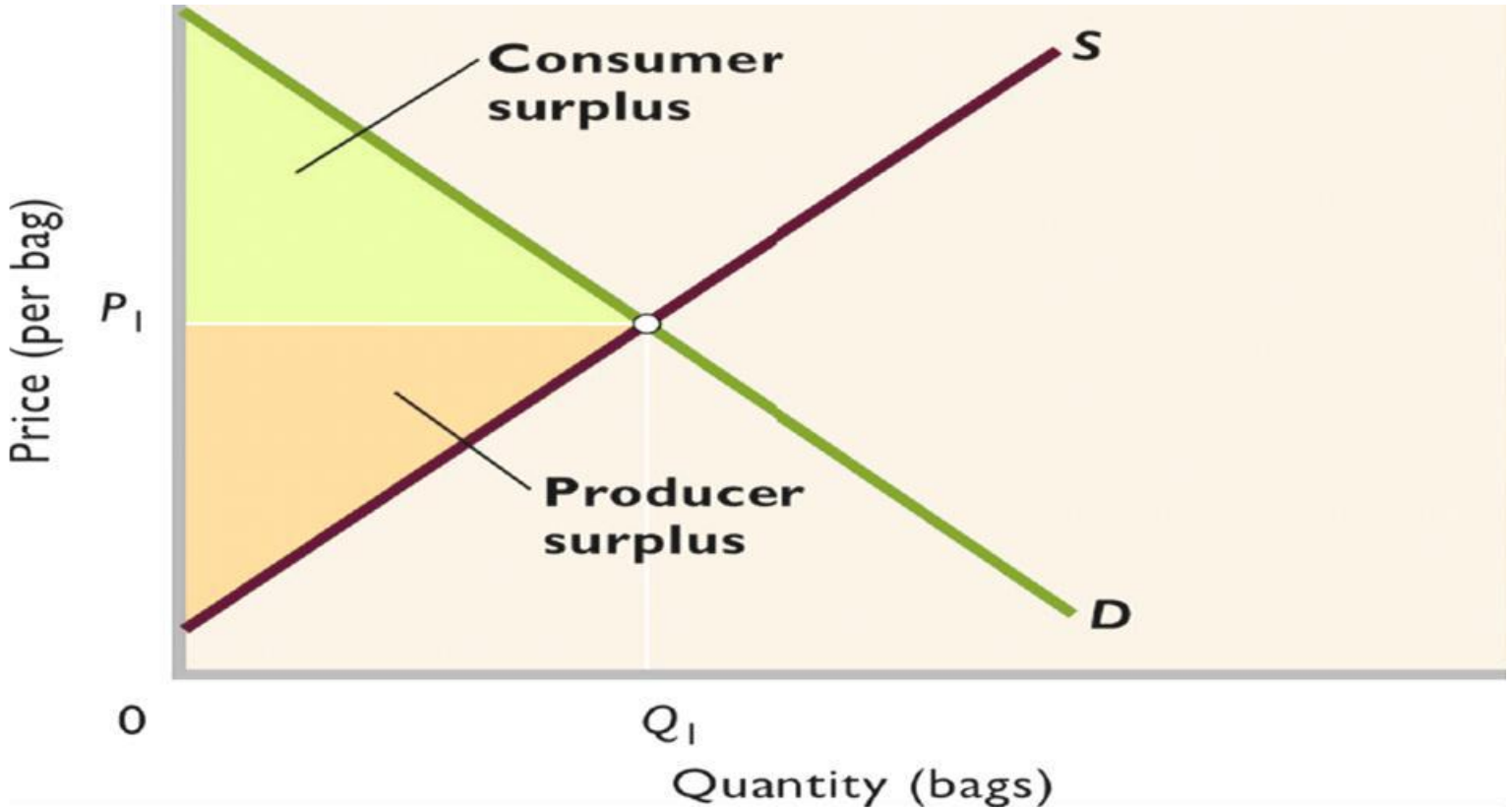
- It is a summation of the individual supply schedules and depicts the supply of different customers for a commodity in relation to its price. Let us understand it with the help of an example.

Price per unit of Banana	Qty supplied by firm A (QA)	Qty supplied by firm B (QB)	Market Supply QA + QB
100	40	50	40 + 50 =90
200	60	70	60 + 70 = 130
300	65	80	65 + 80 = 145
400	80	100	80 + 100 = 180

Consumer surplus & Producer surplus

- Consumer surplus is the difference that the consumer has to pay for a good or service and the price they would be willing to pay rather than forego that good or service.
- Producer surplus is the difference between how much a supplier would be willing to receive for a given quantity of good or service and how much they can actually receive for that quantity based on the market price.

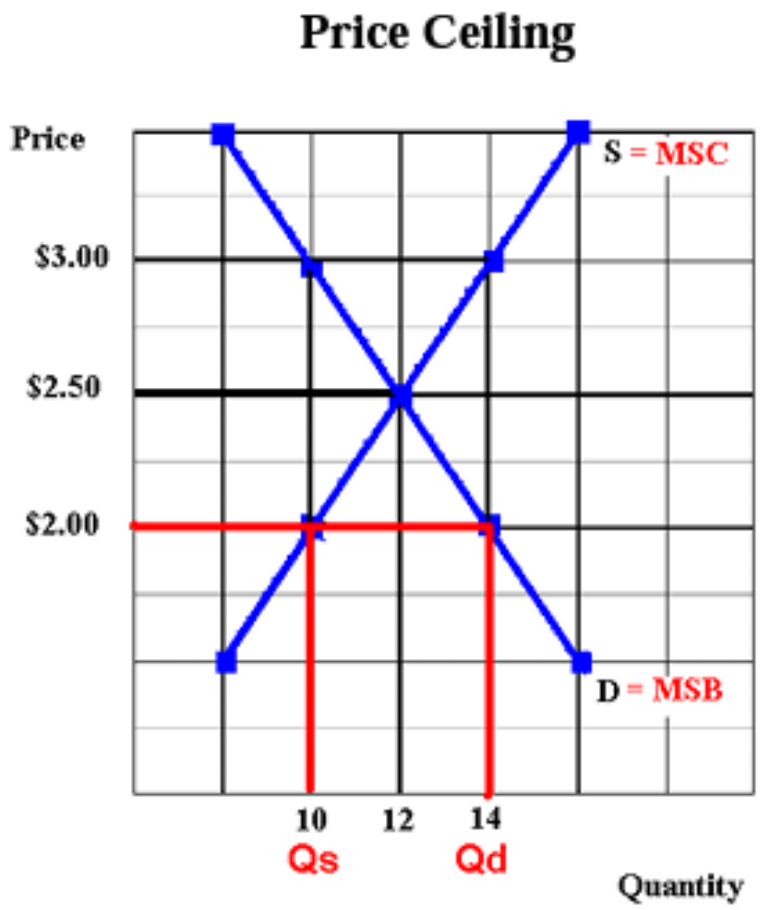
Consumer surplus & Producer surplus

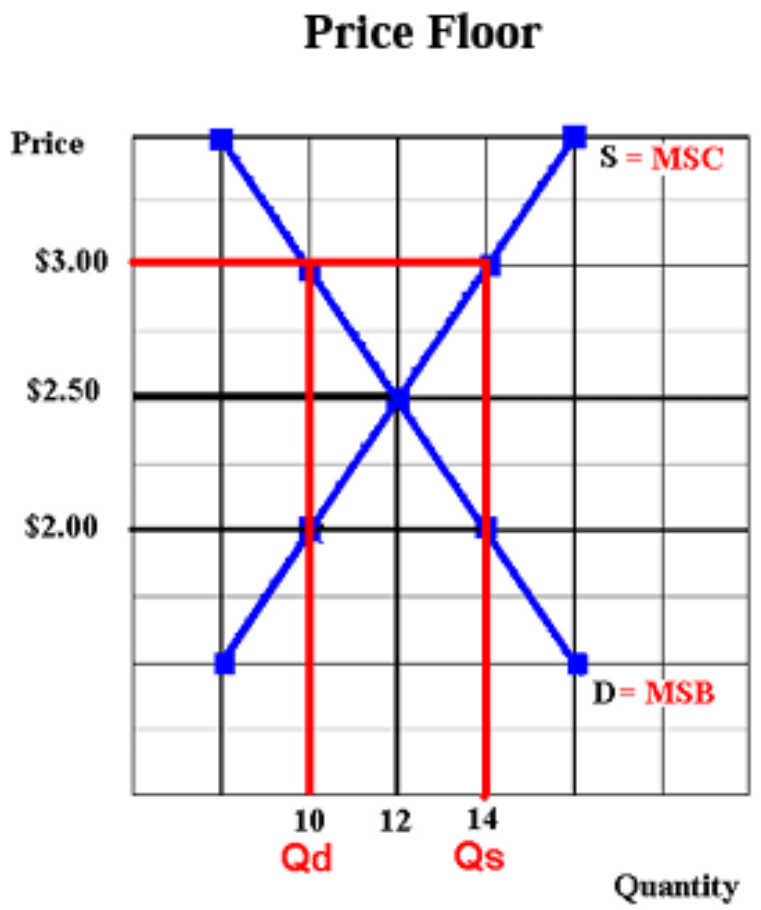


Price Floor and Ceiling

- Laws that government enacts to regulate prices are called Price controls. Price controls come in two flavors. A price ceiling keeps a price from rising above a certain level (the “ceiling”), while a price floor keeps a price from falling below a certain level (the “floor”).
- **Price ceiling:** is a maximum price that can be charged for a product or service. A price ceiling is non-binding when it doesn't have an effect on the market price. A non-binding price ceiling occurs when the price level is greater than or equal to the market equilibrium price. For example, rent control and gasoline.
- **Price floor:** is a minimum price at which a product or service is permitted to sell. A price floor is non-binding when it doesn't have an effect on the market price. A non-binding price floor occurs when the price level is less than or equal to the market equilibrium price. For example, agricultural products and minimum wage

Price Ceiling & Price Floor





At Market Equilibrium

- Price = $2.50 and Quantity = 12
- The point is also the allocative efficient quantity because at Quantity = 12, Marginal Social Benefit = Marginal Social Cost
- **Price Ceiling**
- Ceiling Price = $ 2.00 Quantity Demand = 14 while Quantity Supply 10
- Therefore, the shortage is equal to 4 and this shows that, there is an allocative inefficiency: at the quantity of 10, Marginal Social Benefit > Marginal Social Cost, so there is an **under** allocation of resources.

Price Floor

- Floor Price = $ 3.00 Quantity Demand = 10 while Quantity Supply 14
- Therefore, the surplus is equal to 4 and this shows that, there is an more efficiency: at the quantity of 14,
- Marginal Social Benefit < Marginal Social Cost,
- so there is an **over** allocation of resources.